EVIDENCE MODEL

AI Data Centre Facilities Assurance Framework

Version 1.1

A normative evidence architecture for traceable, current, sufficient and reviewable AI infrastructure assurance

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Evidence is the currency of assurance. Assertions, scores and decisions have no independent authority beyond their evidence, scope and review.

Document control

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Depends on	AIDCF-001 Manifesto; AIDCF-002 Core Conceptual Model; AIDCF-003; AIDCF-004; AIDCF-005; AIDCF-006; AIDCF-007A
Purpose	Define evidence objects, classes, metadata, provenance, integrity, quality, timeliness, coverage, sufficiency, confidence, conflict, UNKNOWN typing and severity, retention and decision use.
Applies to	All AIDCF-007A controls, assertions, Findings, critical gates, tests, scores, maturity determinations, reports, decisions and compatible tooling.
Downstream	AIDCF-009 Assessment Methodology; AIDCF-010 Scoring and Decision Framework; AIDCF-012 Reporting and Assurance Statement Standard; AIDCF-014 and AIDCF-016 machine-executable models; AIDCF-017 Capability Assurance Maturity Model.
Safety/privacy boundary	Evidence collection shall be authorised, proportionate, lawful, minimally intrusive and protected. This standard does not authorise hazardous testing or unrestricted data collection.

Revision history

Version	Date	Summary
1.0	—	Controlled-review draft: full evidence standard and 360-control profile approach.

Version	Date	Summary
1.1	2026-10-10	Live references moved to AIDCF-007A; conformance levels renamed EC0–EC5; quality dimensions aligned to canonical P/F/C/R/T/I meanings; confidence flags; UNKNOWN severity US1–US5 and per-type grading; Evidence Requirement Profiles carried in the AIDCF-007A registry; gate names aligned; consecutive EVM-REQ IDs with mapping annex.

Executive thesis

AIDCF assurance depends on whether a precise assertion is supported by evidence that is relevant, authentic, sufficiently complete, current for the represented state, traceable to its source, protected against inappropriate alteration, and reviewed by a competent accountable party. Evidence is not graded once for all purposes. Its usefulness is evaluated against the claim, Capability Instance, Service Outcome, boundary, operating state, period and decision question.

Canonical evidence chain

Within the canonical sixteen-step assurance chain (AIDCF-002; Decision Question → … → Assurance Decision), this standard governs steps 10 to 13 (Evidence Requirement, Evidence Object, Control Assessment, Finding / UNKNOWN Record) and the evidence inputs to step 14 (Gate Result). The evidence-specific reasoning inside those steps is:

Decision Question -> Assertion -> Evidence Requirement -> Evidence Object -> Provenance and Integrity -> Fitness and Coverage -> Quality and Timeliness -> Sufficiency -> Confidence -> Conflict and UNKNOWN (type and severity) -> Finding / Gate Result -> Bounded Assurance Decision

Key outputs

- Canonical EvidenceObject, EvidencePackage and EvidencePortfolio structures.
- Seven evidence classes (E1–E7) and controlled subtypes.
- Six-dimensional evidence quality model (P, F, C, R, T, I) and quality bands Q0–Q5.
- Timeliness, retention and chain-of-custody rules.
- Assertion-specific sufficiency (S0–S5) and confidence (CONF-0 to CONF-4, flags CONFLICTING and STALE).
- Evidence Requirement Profiles for all 360 controls, carried in the AIDCF-007A registry.
- Critical-gate evidence profiles for CG-01 to CG-15.
- Conflict, derived evidence, automation, UNKNOWN type (U1–U10) and UNKNOWN severity (US1–US5) rules.
- Machine-readable ontology, schema and validation expectations.

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Part I - Authority, scope and conformance

1. Purpose

This standard operationalises the evidence commitments in AIDCF-006 and the Evidence Requirement Profiles carried in the AIDCF-007A Canonical Master Control Registry. It defines a common evidence language suitable for human assessments, structured workpapers, graph implementations and future assurance schemes. Normative keywords (shall, should, may, can) follow ISO/IEC Directives Part 2 usage.

2. Normative requirements

EVM-REQ-001 Every material assertion shall link to supporting, challenging or qualifying Evidence Objects, or to an explicit UNKNOWN Record.

EVM-REQ-002 Evidence shall retain identity, source, collection, scope, time, integrity, custody, classification and review metadata.

EVM-REQ-003 Evidence quality and sufficiency shall be evaluated against a precise assertion; no evidence item has a universal grade.

EVM-REQ-004 DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS and RESILIENCE claims shall use evidence appropriate to the claim type (§29).

EVM-REQ-005 Evidence conflict, staleness, inaccessible sources and coverage gaps shall remain visible.

EVM-REQ-006 UNKNOWN shall not be transformed into pass, zero, false, NOT_APPLICABLE or accepted risk.

EVM-REQ-007 Critical-gate evidence shall meet the declared evidence floor (Part XIII) or the gate shall remain UNKNOWN or FAIL according to the gate criterion and the roll-up rule in AIDCF-010.

EVM-REQ-008 Derived and AI-generated evidence shall retain source lineage, transformation logic and candidate review state.

EVM-REQ-009 Evidence collection and testing shall be authorised, competent, proportionate, lawful and safe.

EVM-REQ-010 Retention and deletion shall preserve legal, contractual, audit and decision-lineage obligations.

EVM-REQ-011 Evidence reuse across controls or methods shall require assertion-specific relevance and scope evaluation.

EVM-REQ-012 A score, maturity level or certificate shall not increase the evidential strength of its underlying sources.

3. Scope

- Evidence created or collected for all 360 AIDCF-007A controls.
- Evidence supporting infrastructure, dependency, failure, safeguard, recovery and graph assertions.
- Evidence needed to evaluate the 15 baseline critical gates (CG-01 to CG-15).
- Evidence used by AIDCF and AI Trust Graph integrated decision packages (AIDCF-005).

- Human, machine, documentary, observational, test, operational and derived evidence.

4. Evidence-system conformance levels

These levels describe how completely an implementation (repository, tool or assessment practice) realises this standard. They were labelled E0–E5 in v1.0 and are renamed EC0–EC5 to avoid confusion with evidence classes E1–E7 (Part IV).

Level	Name	Minimum expectation
EC0	Vocabulary aligned	Uses canonical evidence terms and source ownership.
EC1	Metadata complete	Implements mandatory EvidenceObject fields (§11).
EC2	Provenance controlled	Adds source, collection, integrity, custody and classification (Part VI).
EC3	Assertion linked	Evaluates fitness, quality, timeliness and coverage for precise assertions.
EC4	Decision ready	Supports sufficiency, confidence, gates, conflict, UNKNOWN type and severity, and bounded decisions.
EC5	Machine conformant	Passes schemas, validation shapes, versioning and audit tests (Part XIX).

EVM-REQ-013 An implementation claiming conformance to this standard shall declare its EC level, and each level shall include all requirements of the levels below it. Evidence used for a Gate Result shall be managed by an implementation at EC4 or above.

Part II - Evidence doctrine and mandatory distinctions

5. Foundational doctrines

Doctrine	Rule
Evidence before conclusion	No positive assurance conclusion without affirmative support.
Assertion specificity	Evidence supports or challenges a precise proposition, not an object in the abstract.
Reality over intent	Observed and tested evidence takes precedence where design intent conflicts.
Bounded use	Evidence applies only to stated scope, period, state, conditions and method.
Independent review	Collection, interpretation and decision authority remain distinguishable.
Minimum false precision	Scores express structured judgment but do not manufacture certainty.
Preserved uncertainty	Missing, inaccessible, stale, conflicting or weak evidence remains visible.
Non-inheritance	Third-party certification or prior review is evidence, not automatic assurance.

6. Mandatory distinctions

Do not collapse	Reason
Evidence Object / Evidence Package	One item differs from a governed collection.
Source / Collector / Custodian / Reviewer	Origin, acquisition, storage and judgment are different roles.
Observation time / collection time / record time	The represented state may differ from acquisition or repository entry.

Do not collapse	Reason
Authenticity / integrity	Genuine origin differs from freedom from unauthorised alteration.
Reliability / fitness	A reliable source may not be relevant to, or fit for, the claim.
Coverage / volume	Many records may still cover a narrow population or state.
Quality / sufficiency	Strong evidence can still be insufficient in quantity or diversity.
Evidence quality / confidence	Evidence characteristics differ from reviewer confidence in an assertion.
UNKNOWN type / UNKNOWN severity	Why something is unknown (U1–U10) differs from what it could change (US1–US5).
Stale / contradicting	Old evidence differs from current disagreement.
Independent / external	External does not necessarily mean impartial, competent or verified.
Test result / effectiveness conclusion	A test output requires scope and interpretation.
Derived evidence / source evidence	Analysis does not replace its lineage sources.
Implementation Maturity / control result	Diagnostic maturity (IM0–IM5, AIDCF-017) never substitutes for a control result.

Part III - Evidence metamodel and object hierarchy

7. Core evidence entities

Entity	Definition
EvidenceObject	Smallest independently identified evidence item used to support, challenge or qualify an assertion.
EvidenceSet	A purpose-specific selection of Evidence Objects used together for one assertion or test.
EvidencePackage	A governed collection addressing a control, function, system, scenario or gate.
EvidencePortfolio	A capability-level collection of Evidence Packages and decision records.
EvidenceSource	Person, system, device, provider, repository or process from which evidence originates.
CollectionActivity	Authorised action that acquires or observes evidence.
TransformationActivity	Declared process deriving one evidence object from others.
EvidenceRequirement	Claim-specific statement of required classes, quality, coverage, age and independence; for controls, instantiated from the Evidence Requirement Profile in the AIDCF-007A registry.
EvidenceAssessment	Recorded evaluation of evidence against a requirement.
EvidenceManifest	Machine-readable inventory with identifiers, hashes, locations and status.
CustodyEvent	Transfer, access, export, alteration, redaction or retention action.
EvidenceConflict	Structured disagreement between evidence-supported assertions.
UnknownRecord	Typed (U1–U10) and severity-graded (US1–US5) record of an unresolved evidence condition.

8. Hierarchy

EvidencePortfolio -> EvidencePackage -> EvidenceSet -> EvidenceObject -> Source and Collection Activity -> Assertion / Control Assessment / Gate Result / Assurance Decision

9. EvidenceObject lifecycle states

State	Meaning
EXPECTED	Evidence requirement identified but object not yet collected.
REQUESTED	Collection request issued.
COLLECTED	Object acquired but not quality reviewed.
VALIDATED	Identity, provenance, integrity and readability reviewed.
ACCEPTED	Approved for one or more stated assurance uses.
CHALLENGED	Material concern or contradiction recorded.
STALE	No longer sufficiently representative.
RESTRICTED	Use limited by security, privacy, contract or legal hold.
SUPERSEDED	Replaced by a later object while history remains.
REJECTED	Not suitable for the stated use; rationale retained.
RETIRED	No longer current but retained according to policy.
DESTROYED	Disposed under authorised retention process; destruction record remains.

10. Evidence Package minimum structure

Component	Requirement
Package ID and purpose	Stable identity and decision/claim addressed.
Subject and boundary	Capability Instance, control, system, gate, operating state and period.
Requirement profile	Required evidence classes, quality floor, sufficiency floor, state and period coverage, and independence (from the AIDCF-007A Evidence Requirement Profile or Part XIII).
Manifest	Complete list of included, missing, substituted and restricted items.
Assessments	Per-object quality vector and package sufficiency results.
Conflicts and UNKNOWN	Unresolved contradictions, gaps, UNKNOWN type and severity, and decision effects.
Review and approval	Collector, analyst, reviewer and decision roles.
Version and custody	Immutable release, hash, access and supersession lineage.

Part IV - Evidence taxonomy and subtype catalogue

Class	Name	Representative subtypes
E1	Documentary and declarative	policy; standard; procedure; runbook; design basis; architecture; drawing; contract; attestation; inventory
E2	Configuration and technical state	device configuration; firmware inventory; cluster manifest; access policy; protection setting; topology export; configuration snapshot
E3	Operational and observational	telemetry; log; alarm; event; ticket; work order; capacity metric; performance metric; physical observation; inspection record

Class	Name	Representative subtypes
E4	Validation and test	commissioning; load test; failover test; recovery test; simulation; scenario exercise; functional test; calibration; witnessed demonstration
E5	Independent and third-party	independent engineering review; audit; provider assurance; witnessed test; laboratory result; accredited certificate; regulatory inspection
E6	Human and experiential	interview; workshop; walkthrough; operator demonstration; expert judgment; shift observation; competency evidence
E7	Derived and analytical	trend; correlation; capacity analysis; failure analysis; dependency path; reconciled inventory; score; model output; AI-assisted extraction

Class	Primary claim supported	Use rule
E1 Documentary and declarative	Intent, governance and approved design	Best suited to intent, requirements, authority, design and declared process. It shall not alone prove current implementation or effectiveness.
E2 Configuration and technical state	Implementation and technical state	Requires provenance, extraction context and reconciliation to the current architecture.
E3 Operational and observational	Operation, performance and incidents	Best suited to operation over a stated period. Coverage, completeness, clock integrity, gaps and collection health shall be assessed.
E4 Validation and test	Validation, effectiveness and resilience	Best suited to effectiveness and resilience under stated conditions. Test authorisation, initial state, method, expected result and anomalies shall be retained.
E5 Independent and third-party	Independent corroboration	May strengthen independence but shall be evaluated for scope, competence, method, age, exclusions and transferability.
E6 Human and experiential	Practice, competence and context	Useful for context, operating practice and explanation. Corroboration is required for material technical-effectiveness assertions.
E7 Derived and analytical	Analysis and relationship synthesis	Useful for synthesis and decision support. Derived outputs never exceed the quality, scope and lineage of their sources.

Part V - Evidence metadata standard

11. Mandatory metadata

Field	Requirement
evidenceId	Immutable unique identifier.
title	Concise human-readable name.
description	What the item contains and why it exists.
evidenceClass/subtype	Controlled E1–E7 classification and Appendix A subtype.
sourceId/sourceType	Originating person, system, provider or process.
owner/custodian	Accountable owner and storage custodian.
collector/collectionMethod	Who or what collected it, how and under which authority; method recorded using the canonical assessment-method codes where applicable (INQUIRY, DOCUMENT_REVIEW, INSPECTION, OBSERVATION, RECONCILIATION, ANALYSIS, SAMPLING, TEST, REPERFORMANCE, SCENARIO_EXERCISE).
observationTime	When the represented condition existed.

Field	Requirement
collectionTime	When the evidence was acquired.
recordTime	When entered into the repository.
effectivePeriod	Period for which the evidence is applicable.
subjectRefs	Capabilities, assets, controls, gates, assertions or paths addressed.
scope	Locations, systems, tenant, population and operating states covered (canonical operating states: NORMAL, DEGRADED, MAINTENANCE, EMERGENCY, RECOVERY, TRANSITION, SHUTDOWN, FAILED, UNKNOWN).
architecture/configurationVersion	Baseline represented.
integrityMethod/hash	Integrity protection and digest where applicable.
format/version	File, record, schema and tool versions.
classification	Security/privacy/contractual handling category (§41).
retentionRule	Retention class (§21), trigger and legal hold.
qualityAssessment	Dimension ratings P/F/C/R/T/I and rationale.
reviewState/reviewer	Lifecycle state (§9) and accountable reviewer.
limitations	Known gaps, exclusions, sampling or interpretation limits.
relatedEvidence	Sources, derivatives, conflicts, replacements and package membership.

12. Conditional metadata

- Test evidence: initial state, expected result, procedure, authorisation, abort criteria, instrumentation, anomalies and witnesses.
- Telemetry/log evidence: source clock, timezone, sampling interval, collection health, dropped data and query/filter.
- Configuration evidence: extraction command or interface, privilege used, configuration scope, redactions and target state.
- Human evidence: role, competence, interview protocol, date, corroboration and consent/recording status.
- External evidence: issuer, competence/accreditation where relevant, scope, validity, exclusions and permitted reliance.
- Derived evidence: source IDs, transformation code/query/version, assumptions, quality propagation and reviewer.

Part VI - Provenance, integrity and chain of custody

13. Provenance model

The provenance terms below follow the W3C PROV data model (PROV-DM) at concept level.

Provenance object	Minimum content
Entity	Evidence object or source record.

Provenance object	Minimum content
Activity	Collection, observation, transformation, review or custody event.
Agent	Person, system, provider or organisation responsible.
wasGeneratedBy	Activity that produced the object.
wasDerivedFrom	Source evidence and transformations.
wasAttributedTo	Responsible source or owner.
wasAssociatedWith	Collector, reviewer or tool involved.
used	Source objects consumed by a transformation.
invalidatedAtTime	When evidence became stale, superseded, withdrawn or destroyed.

14. Integrity levels

Level	Name	Anchor
I0	Unknown	Integrity method absent or unverified.
I1	Basic controlled	Repository permissions and version history.
I2	Verified copy	Source comparison or controlled export metadata.
I3	Hashed	Cryptographic digest recorded in manifest.
I4	Signed / trusted timestamp	Digital signature or trusted timestamp validated.
I5	Tamper-evident custody	End-to-end signed custody, immutable storage or equivalent control.

The integrity level of an object and its I quality rating (§17) use the same anchors; the I rating records the level verified for the specific use.

15. Chain-of-custody events

- Creation or acquisition.
- Transfer between collectors, custodians or repositories.
- Read/access for review.
- Transformation, parsing, redaction or export.
- Integrity verification.
- Challenge, correction or supersession.
- Legal hold, restriction, retention extension or authorised destruction.

EVM-REQ-014 Every material transformation shall produce a new Evidence Object linked to source evidence; original evidence shall not be silently overwritten.

EVM-REQ-015 Redaction shall preserve an unredacted controlled source where lawful and shall record the basis, method and affected content.

Part VII - Evidence quality dimensions and grading

16. Quality dimensions

Code	Dimension	Question
P	Provenance	Can origin, authority and collection lineage be established?
F	Fitness (relevance)	Does the item address the precise subject, criterion, claim type, configuration and state of the assertion?
C	Coverage	Does it cover the required population, period, state and boundary?
R	Reliability	Is the method/source repeatable, controlled and corroborated?
T	Timeliness (currentness)	Is the item sufficiently current and representative of the state being asserted?
I	Integrity	Is it protected against undetected alteration and custody loss?

Alignment note (v1.1): in v1.0, F denoted Freshness and T denoted Traceability. The canonical meanings are now F = Fitness/relevance and T = Timeliness/currentness; the former Freshness anchors are carried unchanged as the T anchors. Traceability is no longer rated per item because it is a property of the assessment record rather than of the evidence; it is enforced as a mandatory linkage requirement (§17A) and by validation rules EVM-VAL-002 and EVM-VAL-006.

17. Rating scale

P rating anchors

Rating	Anchor
P0	Unknown origin
P1	Identified but uncontrolled source
P2	Accountable internal source
P3	Controlled authoritative source
P4	Corroborated authoritative source
P5	Independently verified origin and lineage

F rating anchors

Rating	Anchor
F0	Relevance not established, or the item is not linked to the assertion
F1	Indirect: concerns a related system, population or design generation, not the asserted subject
F2	Partial: correct subject but a different claim type, configuration version or operating state
F3	Relevant: addresses the asserted subject, claim type and period, and is linked to the assertion
F4	Specific: addresses the precise assessment criterion, configuration version and declared operating states
F5	Discriminating: designed for the criterion (e.g. a test with pre-declared acceptance criteria) so that a nonconforming condition would have been detected

C rating anchors

Rating	Anchor
C0	Unknown coverage
C1	Anecdotal or isolated
C2	Limited sample/state
C3	Representative core scope
C4	Broad scope and relevant states
C5	Comprehensive declared population, period and states

R rating anchors

Rating	Anchor
R0	Unassessed
R1	Weak/uncontrolled method
R2	Plausible but lightly corroborated
R3	Controlled repeatable method
R4	Strong and independently corroborated
R5	Validated method and independent verification

T rating anchors

Rating	Anchor
T0	Unknown age or state
T1	Obsolete for represented state
T2	Stale or change-sensitive
T3	Acceptable for decision
T4	Current and aligned to baseline
T5	Near-real-time or event-current with healthy collection

I rating anchors

Rating	Anchor
I0	Unknown integrity
I1	Basic repository controls
I2	Verified source copy
I3	Hash verified
I4	Signature/timestamp verified
I5	Tamper-evident end-to-end custody

17A. Linkage requirement (formerly the T Traceability dimension)

EVM-REQ-016 Every ACCEPTED Evidence Object used for a control result shall be linked to its subject, the assertion it supports, challenges or qualifies, and the period represented. An object without this linkage shall be rated F0 for that use.

EVM-REQ-017 Evidence used for a Gate Result shall additionally be linked end to end to the Finding or UNKNOWN Record and to the Gate Result it informs; evidence used for an Assurance Decision shall be reachable from the decision through an immutable EvidenceManifest (EVM-VAL-015).

18. Evidence quality vector

Preferred expression: [Class]-P#-F#-C#-R#-T#-I#. Example: E4-P4-F4-C3-R4-T4-I3. The vector shall remain visible. A single averaged number may be used only as a secondary sorting aid and shall not replace the dimensions. The only sanctioned averaged number is the Evidence Quality Index (EQI) defined in AIDCF-010; it never changes a quality band, a sufficiency level, a control result, a Gate Result or a decision.

EVM-REQ-018 Quality floors in Evidence Requirement Profiles shall be expressed per dimension (for example "P3 F3 C3 R3 T3 I2") and shall be tested per dimension; an item that misses any dimension of the floor does not meet the floor regardless of its EQI.

19. Evidence quality bands

Quality bands Q0–Q5 replace the "evidence grade E0–E5" used in AIDCF-005 §37 and AIDCF-006 §11.

Band	Rule	Permitted use
Q0 Unusable	Any dimension P0, F0, R0, T0 or I0 without accepted limitation.	Context only; no positive assertion.
Q1 Weak	Multiple dimensions at 0–1 or unsupported source.	Candidate/UNKNOWN; remediation priority.
Q2 Limited	Core provenance established but limited coverage, timeliness or corroboration.	Descriptive or limited DESIGN claim.
Q3 Adequate	All claim-critical dimensions at least 3.	DESIGN or IMPLEMENTATION claim where appropriate.
Q4 Strong	Most dimensions 4+, with no claim-critical dimension below 3.	OPERATION or EFFECTIVENESS claim with sufficient package coverage.
Q5 Highly defensible	Independent/corroborated, representative, current and fit for the criterion, with strong integrity.	High-consequence gate or RESILIENCE claim, subject to method and sufficiency.

Part VIII - Timeliness, validity, retention and legal hold

20. Timeliness rule

Timeliness (dimension T) is risk- and change-based. The windows below are default review triggers, not universal expiry. A material change invalidates representativeness even when the nominal window has not elapsed.

Evidence category	Default currency basis	Immediate invalidation/review triggers
Live telemetry and active alarms	Current collection health; decision-specific lookback	Collection outage, sensor/configuration change, incident.
Event and security logs	Usually current period and sufficient lookback for decision	Clock/source/query change, incident, retention gap.
Configuration snapshots	At assessment and after material change	Firmware, policy, topology, identity or setpoint change.
Capacity/performance metrics	Current demand cycle and peak/seasonal state	Expansion, workload shift, throttling or demand change.
Maintenance/calibration records	Current maintenance cycle	Overdue work, failure, deferral or scope change.
Commissioning and validation tests	Until material change, adverse event or review trigger	Architecture, load, firmware, procedure or provider change.
Recovery and continuity exercises	Risk-based periodic cycle and after material change	Failure, failed exercise, new dependency or recovery design change.
Architecture/design documents	Until superseded or material change	As-built discrepancy, expansion or reconfiguration.
Contracts/provider attestations	Until expiry, renewal or service change	Provider, SLA, subprocessor or jurisdiction change.
Interviews and walkthroughs	Point-in-time context	Role/process change or contradictory technical evidence.

21. Retention classes

Class	Purpose	Rule
R0 Transient	Temporary extraction or working copy.	Delete under controlled process after verified ingestion unless needed for decision traceability.
R1 Operational	Routine monitoring and work records.	Retain for operational, contractual and investigation needs.
R2 Assurance	Assessment evidence and workpapers.	Retain through decision validity plus defined challenge/reassessment period.
R3 Critical decision	Gate, resilience and high-consequence evidence.	Longer risk-based retention with immutable lineage.
R4 Legal/regulatory hold	Evidence subject to preservation obligation.	No alteration/destruction until authorised release.
R5 Canonical methodology	Schemas, mappings, approved examples and release artifacts.	Versioned permanent archive.

Retention class codes R0–R5 are distinct from reliability ratings R0–R5 (§17); records shall qualify them (retentionClass versus quality.R).

EVM-REQ-019 Retention periods shall be defined by applicable legal, contractual, regulatory, security, privacy and operational requirements; this standard does not prescribe universal years.

Part IX - Coverage, sampling and representativeness

22. Coverage dimensions

Dimension	Coverage question
Entity	Which assets, systems, providers and interfaces are included?
Population	What proportion and which strata of the population are represented?
Location	Which sites, halls, zones, racks and jurisdictions?
Time	Which period, season, demand cycle and incident window?
Operating state	Which of NORMAL, DEGRADED, MAINTENANCE, EMERGENCY, RECOVERY, TRANSITION, SHUTDOWN and FAILED are represented?
Scenario	Which threat, hazard, failure, common-mode and recovery paths?
Tenant/workload	Which tenancy, cluster, workload or Service Outcomes?
Control behaviour	Design, implementation, operation, monitoring, validation and recovery?
Dependency	Which upstream, downstream, shared and external dependencies?

23. Sampling principles

- Sampling shall be risk-based, reproducible and documented.
- The population, strata, selection method, sample size rationale and exclusions shall be recorded.
- Judgmental sampling may target critical or anomalous items but shall not be represented as statistical coverage.
- Random or systematic sampling requires a trustworthy population frame.
- A sampled pass shall not be extrapolated beyond the represented population and conditions.
- Exceptions and failed samples shall influence expansion and conclusion.

24. Coverage rating

Level	Meaning
C0	Population or scope unknown.
C1	Single example or anecdotal evidence.
C2	Limited sample or one state.
C3	Representative core population and expected state.
C4	Broad population and material DEGRADED/MAINTENANCE states.
C5	Comprehensive declared population, periods and applicable states, subject to practical limits.

Evidence coverage (C, per item or package) is distinct from assessment coverage (the percentage of in-scope controls with a scored result), which is defined in AIDCF-010.

Part X - Evidence sufficiency and confidence

25. Sufficiency model

Sufficiency is an assertion-specific judgment based on whether the combined evidence package meets required class diversity, quality floors, coverage, timeliness, independence, conflict resolution and state/scenario representation. It shall not be calculated only from evidence count.

Level	Name	Meaning
S0	None	No usable evidence.
S1	Insufficient	Some evidence exists but cannot support the claim.
S2	Minimum descriptive	Supports a bounded description or design intent.
S3	Reasonable	Supports DESIGN or IMPLEMENTATION within declared scope.
S4	Strong	Supports OPERATION or EFFECTIVENESS for representative conditions.
S5	Highly defensible	Supports high-consequence gate or RESILIENCE claim with diverse, current, traceable and appropriately independent evidence.

EVM-REQ-020 Where the sufficiency of the evidence package for a claim is below the sufficiencyFloor of the applicable Evidence Requirement Profile, the control result for that claim shall be UNKNOWN (not PARTIAL or NONCONFORMING), except that a NONCONFORMING result may be recorded on evidence below the floor where that evidence affirmatively demonstrates that a criterion is not met; the rationale shall be recorded.

26. Sufficiency decision factors

Factor	Assessment
Required class diversity	Are design, configuration, operation and validation classes present as required?
Quality floor	Does each mandatory item meet the per-dimension floor?
Complementarity	Do items address different weaknesses rather than duplicate one source?
Independence	Are critical claims corroborated by an appropriately separate source/method?
Coverage	Does the package represent population, state, time and scenario?
Conflict	Are material contradictions resolved or bounded?
UNKNOWN	Are missing elements identified with type, severity and decision effect?
Critical gate	Does the package meet the declared non-compensating gate floor (Part XIII)?

27. Confidence

Confidence is a reviewer's recorded professional judgment of support for a bounded assertion. It has five levels and two flags.

Level	Meaning
CONF-0 UNKNOWN	No defensible confidence state.
CONF-1 LOW	Weak/limited evidence; high sensitivity to assumptions.

Level	Meaning
CONF-2 MODERATE	Reasonable support with material limitations.
CONF-3 HIGH	Strong support with no unresolved material conflict within scope.
CONF-4 VERY_HIGH	Highly defensible, independently corroborated and representative support for the bounded claim.

Flag	Meaning	Effect
CONFLICTING	Material evidence supports incompatible propositions (U4).	Confidence for the affected assertion shall not exceed CONF-1 until the conflict is resolved or bounded under §34.
STALE	Previously supported but no longer sufficiently current (U3).	Confidence for current-state use reverts to CONF-0; historical use may retain its prior level with the period stated.

Flags are recorded alongside a level (for example "CONF-1, CONFLICTING"); they are not levels.

Ceiling source	Maximum confidence
Package sufficiency S0–S1	CONF-0
Package sufficiency S2	CONF-1
Package sufficiency S3	CONF-2
Package sufficiency S4	CONF-3
Package sufficiency S5	CONF-4
An attached US2 UNKNOWN (§35A)	CONF-2
An attached US3, US4 or US5 UNKNOWN	CONF-1 for the affected assertion

These are ceilings, not a formula: a reviewer may record lower confidence with rationale.

EVM-REQ-021 Confidence shall be recorded with rationale and shall not be derived from a hidden formula.

EVM-REQ-022 A confidence level shall not exceed the ceiling imposed by the weakest mandatory evidence requirement, an attached flag or an attached UNKNOWN Record.

Part XI - Evidence-to-assertion and result model

28. Assertion structure

Field	Requirement
assertionId	Stable versioned identity.
proposition	Subject-predicate-object or equivalent precise statement.
claimType	For control assessments, one of DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS or RESILIENCE. Other assertions may use the supplementary kinds in §29 (intent/governance, absence, independent assurance, derived decision).
scope/conditions	Capability Instance, boundary, operating state, configuration, load, period and assumptions.
evidenceRequirement	Required classes, quality floors, sufficiency floor, coverage and independence.
supportingEvidence	Evidence Objects supporting proposition.
challengingEvidence	Evidence Objects contradicting or weakening proposition.
qualifyingEvidence	Evidence limiting scope or interpretation.

Field	Requirement
assessment	Quality, sufficiency, conflict and UNKNOWN evaluation.
confidence/review	Confidence level and flags, reviewer and date.
decisionUse	Findings, gate or decision that relies upon the assertion.

29. Claim-type evidence floors

Claim type	Minimum Assurance Level	Minimum pattern
DESIGN	AL2	E1 design evidence plus competent technical analysis; E5 where required by authority or consequence.
IMPLEMENTATION	AL3	E2 configuration/as-built evidence plus E3 or E6 observation; reconciled to the approved design.
OPERATION	AL4	E3 over a representative period and the declared operating states, plus current E2 and exception/incident review.
EFFECTIVENESS	AL5	E4 representative validation under tested or observed adverse conditions, plus E2/E3 supporting state and operating evidence.
RESILIENCE	AL5	E4 scenario/recovery evidence plus cross-domain E2/E3 and dependency coverage, demonstrating containment and recovery within tolerance.

Supplementary assertion kind	Minimum pattern
Intent/governance	E1 plus accountable approval and current applicability.
Negative/absence claim	Documented discovery coverage, method sensitivity and residual UNKNOWN.
Independent assurance	E5 scope and method plus access to underlying evidence where needed.
Derived decision	E7 with complete source lineage and human review.

30. Control result and finding model

Control results use the canonical codes defined in AIDCF-010 (per control, per claim type, per scope). Their evidence rules are:

Result / record	Evidence rule
CONFORMING	Affirmative evidence meets every assessment criterion for the claim type and the package meets the profile floors.
PARTIAL	Some criteria are met on evidence at or above floor; each unmet criterion is recorded as a Finding.
NONCONFORMING	Evidence demonstrates that criteria are materially not met, or that a safeguard is ineffective.
UNKNOWN	Evidence below floor, inaccessible, stale, conflicting or unresolved; an UNKNOWN Record with type and severity is mandatory.
NOT_APPLICABLE	Approved evidence-based non-applicability rationale and approver.
NOT_ASSESSED	In scope but not yet assessed; no evidence judgement made.
Observation (Finding type)	Evidence describes a condition without determining a control result.
Conflicting evidence	Not a result: recorded as UNKNOWN (U4) with the CONFLICTING confidence flag until resolved.

Result / record	Evidence rule
Suspended reliance	Not a result: the ControlAssessment lifecycle state becomes SUSPENDED and the result reverts to UNKNOWN (U3) until reassessed.

Part XII - Evidence-to-control mapping (Evidence Requirement Profiles)

31. Control Evidence Requirement Profile

Every AIDCF-007A control has an Evidence Requirement Profile. As of this version the profiles for all 360 controls are carried in the AIDCF-007A Canonical Master Control Registry as the evidenceProfile structure of each control record (one per control; per-domain machine files and CSV columns derived from it). This standard defines the meaning and rules of the fields; the registry holds the values. A profile is normative for evidence classes and floors but may be strengthened, never weakened, by a capability profile, criticality, scenario or decision need.

Field	Type / vocabulary	Meaning
claimType	One of DESIGN, IMPLEMENTATION, OPERATION, EFFECTIVENESS, RESILIENCE	The highest claim type the control is designed to support; consistent with the control's requiredAssuranceLevel (DESIGN AL2, IMPLEMENTATION AL3, OPERATION AL4, EFFECTIVENESS and RESILIENCE AL5). Lower claim types are assessed with the patterns in §29.
profileTier	CP-1 to CP-5	Control profile tier (§32).
requiredClasses	Array of E1–E7	Classes that shall be present in the package for the claimType.
optionalClasses	Array of E1–E7	Classes that may strengthen the package or serve as acceptable alternatives.
qualityFloor	String, e.g. "P3 F3 C3 R3 T3 I2"	Minimum rating per dimension for each mandatory item (§18).
sufficiencyFloor	S2 to S5	Minimum package sufficiency for a CONFORMING or PARTIAL result (§25).
stateCoverage	Array of canonical operating states	Operating states the package shall represent.
periodCoverage	String	Required observation period or risk-based window (for example "≥ one full maintenance cycle including a MAINTENANCE state").
independence	String	Required source/method separation (for example "test witnessed by a party independent of the installer").
mandatoryItems	Array of strings	Non-substitutable evidence items.
acceptableAlternatives	Array of strings	Equivalent evidence with the rationale that makes it equivalent.
unknownEffect	String	UNKNOWN severity (US1–US5) to apply when the profile cannot be met, and its effect.
reassessmentTriggers	Array of strings	Conditions that invalidate the evidence and require reassessment.

EVM-REQ-023 The Evidence Requirement Profile applied in an assessment shall be the profile carried for that control in the version of the AIDCF-007A registry declared in the assessment plan; any strengthening shall be recorded with rationale in the Evidence Package, and any weakening shall not be permitted.

EVM-REQ-024 A profile's unknownEffect shall state a severity determined with the test in §35A; where the control holds a CRITICAL gate mapping in the registry, the unknownEffect shall be at least US4 for any claim type

mandatory for that gate.

32. Control profile tiers

Tier	Use	Minimum evidence pattern
CP-1 Foundational	Governance, inventory, design intent.	E1 plus accountable source; S2/S3.
CP-2 Technical state	Configuration, topology and implementation.	E1+E2 with observation/reconciliation; S3.
CP-3 Operational	Operation, monitoring and maintenance.	E2+E3 across representative period; S3/S4.
CP-4 Effectiveness	Controls tested under defined conditions.	E2+E3+E4; S4.
CP-5 Critical/resilience	Critical gate, compound scenario or high-consequence outcome.	Diverse E1–E4 and appropriate E5; S5 or profile-defined strong floor.

Part XIII - Critical-gate evidence profiles

Gate names and scopes are canonical (AIDCF-010 Part VIII, which also defines the gate criteria, roll-up rule and decision ceilings). The machine-readable register is registries/gates/critical-gates.csv with criteria in registries/gates/critical-gate-criteria.csv.

Gate	Name	Mandatory evidence profile	Classes	Floor	Evidence effect (decision ceilings in AIDCF-010)
CG-01	Capability and outcome definition	Capability register; approved Service Outcomes and tolerances; criticality and tolerance analysis; architecture linkage; decision question	E1+E7	S3	An undefined or unapproved Capability Instance or outcome blocks any positive capability assurance.
CG-02	Boundary and architecture integrity	Approved boundary; current architecture; as-built/configuration reconciliation; exclusions and external dependencies	E1+E2+E6	S3	An unclear material boundary or unreconciled architecture blocks a final conclusion for the affected area.
CG-03	Critical dependency visibility	Dependency graph; source/route/provider evidence; common-mode and concentration analysis; ownership	E1+E2+E7	S4	A critical hidden dependency remains UNKNOWN, or FAIL if demonstrated unidentified or unowned.
CG-04	Life safety and hazardous energy	Authorised engineering/safety records; arc-flash and hazard studies; inspection; permits; tests; unresolved hazard status	E1+E4+E5	S4	An unsafe unresolved condition blocks relevant assurance and any hazardous testing in the affected area.
CG-05	Resilient electrical supply	SLD; protection/coordination and capacity study; UPS/BESS/generator tests; fuel; maintenance; EPMS/incident evidence	E1+E2+E3+E4	S5	Missing critical continuity evidence gives UNKNOWN; demonstrated inability gives FAIL.
CG-06	Thermal continuity	Thermal design/model; P&ID; capacity/flow telemetry; leak/transfer/failover tests; maintenance	E1+E2+E3+E4	S5	An unverified critical heat-removal path gives UNKNOWN; demonstrated shortfall gives FAIL.
CG-07	Critical control-system integrity	BMS/EPMS/CDU/PLC/SCADA architecture; configuration and logic; identity/authority; telemetry; fail-safe tests	E1+E2+E3+E4	S4	Uncontrolled or unverifiable critical automation limits or blocks assurance.

Gate	Name	Mandatory evidence profile	Classes	Floor	Evidence effect (decision ceilings in AIDCF-010)
CG-08	Compute, storage and network functional integrity	Current topology/configuration; capacity/performance telemetry; error data; failover/recovery tests	E2+E3+E4	S4	An unverified critical infrastructure function caps OPERATION/RESILIENCE assurance.
CG-09	Failure containment	Fault-domain model; isolation design; scenario analysis; containment tests/incidents	E1+E2+E4+E7	S4	Absent or failed containment may be FAIL; an untested critical boundary is UNKNOWN.
CG-10	Recovery readiness	Recovery plan; prerequisites; configuration/data sources; exercises/tests; defects and sign-off	E1+E2+E3+E4	S5	A plan without representative test supports DESIGN only, not recoverability.
CG-11	Evidence sufficiency	Evidence requirements; manifest; quality/sufficiency assessments; conflicts and limitations	E1+E7	S4	Failure to meet the evidence floor blocks the dependent claim.
CG-12	Critical UNKNOWN	UNKNOWN register with type and severity; consequence; resolution ownership; evidence requests; decision ceiling	E1+E7	S3	An undispositioned US4/US5 UNKNOWN caps or blocks assurance.
CG-13	Responsibility and authority	RACI; delegations; owner/operator/provider clauses; escalation; decision authority	E1+E6	S3	An unowned critical function or dependency blocks accountable disposition.
CG-14	Change control	Baseline; change request; impact analysis; test/rollback; approval; post-change validation	E1+E2+E3+E4	S4	Material uncontrolled change suspends affected evidence and decision.
CG-15	Integrated digital authority over infrastructure	Identity and privilege records; reachability; control-plane interfaces; session logs; action/rollback tests	E2+E3+E4+E7	S4	Unverified consequential authority remains a US4 UNKNOWN.

EVM-REQ-025 Evidence for a critical gate shall not be substituted below its mandatory class or quality floor merely to produce a complete scorecard.

Part XIV - Conflict, contradiction and UNKNOWN

33. Evidence relationship states

State	Meaning
SUPPORTS	Evidence increases support for a precise assertion.
CHALLENGES	Evidence weakens or contradicts assertion.
QUALIFIES	Evidence narrows scope or conditions.
INCOMPLETE	Evidence covers only part of requirement.
STALE	Evidence no longer represents current state.
INACCESSIBLE	Evidence is believed to exist but cannot be reviewed.
UNVERIFIED	Origin, integrity or method is not sufficiently established.
SUPERSEDES	New object replaces current use while preserving history.

State	Meaning
DUPLICATES	Same underlying source is represented more than once.
CONFLICTS_WITH	Two supported propositions are materially incompatible.

SUPPORTS, CHALLENGES, QUALIFIES and SUPERSEDES are canonical graph relationships (AIDCF-014 §9). INCOMPLETE, STALE, INACCESSIBLE, UNVERIFIED, DUPLICATES and CONFLICTS_WITH are evidence-assessment states recorded on the EvidenceAssessment; in the graph, a conflict is expressed as CHALLENGES between the conflicting evidence and assertion plus an EvidenceConflict node.

34. Conflict workflow

Detect -> register separate assertions -> preserve both evidence lineages -> assess scope/time/state differences -> seek corroboration -> competent review -> resolve, bound or escalate -> update decision while preserving history

35. UNKNOWN types

Code	Type	Definition
U1	Missing	Expected evidence not produced.
U2	Inaccessible	Evidence exists but access is restricted or unavailable.
U3	Stale	Evidence does not represent current state.
U4	Conflicting	Material sources disagree.
U5	Coverage gap	Required population, state, period or scenario not represented.
U6	Source unverified	Origin/authority cannot be established.
U7	Integrity uncertain	Alteration or custody cannot be ruled out.
U8	Method uncertain	Collection/test method insufficiently defined.
U9	Interpretation unresolved	Competent reviewers cannot yet determine meaning.
U10	Dependency unknown	Evidence cannot establish a material external/shared dependency.

EVM-REQ-026 Each UNKNOWN Record shall identify subject, missing attribute, type (U1–U10), severity (US1–US5) with the test that determined it, reason, potential consequence, evidence needed, owner if known, status and decision/gate impact.

35A. UNKNOWN severity

Type explains why something is unknown; severity states what it could change. Severity is determined by applying the tests below top-down and assigning the first that applies. The same type can take any severity.

Code	Name	Test (first that applies, top-down)	Decision effect
US5	Systemic	Affects the assessment boundary, a Capability Instance or Service Outcome definition, or two or more gates.	Decision ≤ INCONCLUSIVE.
US4	Critical	Affects a gate-critical control (a control with a CRITICAL mapping for an in-scope gate) or a gate criterion; triggers CG-12.	Gate → UNKNOWN; decision ≤ LIMITED or INCONCLUSIVE per that gate's UNKNOWN ceiling (AIDCF-010).

Code	Name	Test (first that applies, top-down)	Decision effect
US3	Material	Could change a control result or a decision condition, but affects no gate.	Decision ≤ CONDITIONALLY_ASSURED; a condition is recorded.
US2	Moderate	Affects confidence but could not change any result.	Confidence ceiling CONF-2 for affected claims.
US1	Low	No plausible effect on any result or condition.	Record only.

How each UNKNOWN type is graded. The guidance below states what the assessor examines for each type when applying the test; it does not replace the test.

Type	Grading question	Typical outcome
U1 Missing	What does the missing item support? If it is a mandatoryItem in the profile of a gate-critical control, or the evidence for a gate criterion, US4. If it is a mandatoryItem of a non-critical control, US3 (result becomes UNKNOWN). If an alternative item already meets the floor, US2 or US1.	US3; US4 for gate-critical controls; US5 if the missing item is the approved boundary or outcome definition.
U2 Inaccessible	Would access plausibly reveal a different result? Restricted provider evidence for a dependency on which a gate relies (for example a utility or colocation provider's maintenance records) is US4; where accessible corroborating evidence meets the floor, US2.	US3/US4; never downgraded on the assertion of the evidence holder alone.
U3 Stale	Has a material change occurred since observation (§20 triggers)? If yes and the claim concerns current state of a gate-critical control, US4; if no change is evidenced and only the nominal window has lapsed, US2.	US2–US4 depending on change evidence.
U4 Conflicting	Do the conflicting propositions lead to different results? If one proposition would make a gate-critical control NONCONFORMING, US4; if both lead to the same result, US2.	US3/US4 until resolved under §34.
U5 Coverage gap	Is the uncovered population, state or scenario one the Service Outcome tolerance depends on (for example the MAINTENANCE or DEGRADED state for a concurrently maintainable path)? If so and the control is gate-critical, US4; if the gap is the declared operating envelope itself, US5.	US3/US4; US5 where it spans the envelope or ≥ 2 gates.
U6 Source unverified	Is the item decisive? An unverified source for the only evidence of a gate criterion is US4; where corroborated by an independent verified source, US2.	US3/US4.
U7 Integrity uncertain	Could undetected alteration change the result? Integrity doubt over test or configuration evidence for a gate-critical control is US4; a suspected custody breach affecting evidence across several gates is US5 and is also treated as an evidence nonconformance (§47).	US4/US5 for decisive evidence.
U8 Method uncertain	Could the method have failed to detect a nonconforming condition (F below 4 for a test)? If so for a gate-critical claim, US4; otherwise US3.	US3/US4.
U9 Interpretation unresolved	Do the competing interpretations give different results? Grade in the same way as U4.	US2–US4.
U10 Dependency unknown	Does the unknown dependency affect a redundancy, independence or recovery claim? If it could create a common mode across paths relied on by one gate, US4 (CG-03); across two or more gates (for example a shared controller or shared utility path behind both power and cooling), US5.	US4/US5.

EVM-REQ-027 Severity shall be assigned by the assessor using the top-down test, recorded with the test that determined it, and reviewed by the independent reviewer for every US3, US4 and US5 UNKNOWN.

EVM-REQ-028 An UNKNOWN Record's severity shall not be downgraded without new evidence or an approved boundary change, and the prior severity, rationale and approver shall be retained.

Part XV - Derived, automated and AI-generated evidence

36. Derived evidence rules

- Retain every source Evidence ID and the transformation method.
- Record query, script, model, parameters, assumptions and version.
- Do not assign a quality dimension higher than source support without independent corroboration relevant to that dimension.
- Distinguish deterministic transformation from interpretive inference.
- Preserve failed, excluded and conflicting source records where material.
- Recalculate or invalidate derived evidence when a source changes or is withdrawn.

37. AI-assisted evidence

Use	Permitted status	Required safeguards
Classification/extraction	Candidate	Source link, model/version, prompt/process record, review and sampling.
Summarisation	Candidate workpaper	Traceable citations to source passages and reviewer verification.
Anomaly detection	Signal/candidate	Baseline, sensitivity, false-positive/negative review and corroboration.
Path inference	Candidate graph assertion	Approved semantics, source evidence and accountable human review.
Quality scoring	Recommendation	Transparent inputs/rules and override rationale.
Final finding/decision	Not autonomous	Competent accountable human approval required.

EVM-REQ-029 AI-generated text or graph content shall not be treated as independent corroboration of the source data from which it was generated.

Part XVI - Evidence lifecycle and review workflow

38. Lifecycle

Plan -> Request -> Collect -> Identify -> Validate -> Classify -> Protect -> Assess Quality -> Link to Assertion -> Assess Sufficiency -> Review Conflict/UNKNOWN (type and severity) -> Accept or Reject -> Use in Decision -> Monitor Timeliness -> Supersede/Retain/Destroy

39. Role model

Role	Responsibility
Evidence owner	Accountable for source availability, quality and authorised use.
Collector	Acquires evidence under approved method and records provenance.
Custodian	Protects, versions, retains and controls access.

Role	Responsibility
Technical specialist	Evaluates engineering or technical meaning.
Assessor	Links evidence to criteria and assertions; assigns UNKNOWN type and severity.
Independent reviewer	Challenges sufficiency, conflict, UNKNOWN severity and decision use.
Decision owner	Authorises bounded conclusion and conditions.
Privacy/security/legal reviewer	Determines handling, redaction, disclosure and hold obligations.
Tool/system owner	Maintains collection and transformation integrity.

40. Evidence request record

Field	Content
requestId	Stable identifier.
decision/assertion/control	Why evidence is required.
requested class/floor	Classes, metadata, quality, period, state and format.
source/owner	Expected provider and accountable owner.
authority/purpose	Permitted collection and use.
due/review trigger	Operational scheduling field.
received/status	Collected, partial, inaccessible, rejected or UNKNOWN.
handling	Classification, transfer and retention.
substitution	Alternative evidence and approval rationale.
escalation	Decision and gate effect of non-production, including the UNKNOWN severity that will apply.

Part XVII - Evidence security, privacy and disclosure

41. Classification levels

Level	Examples	Handling intent
PUBLIC	Published methodology, synthetic examples.	Approved public release.
INTERNAL	General workpapers and non-sensitive procedures.	Authenticated organisational access.
CONFIDENTIAL	Client architecture, contracts, findings and operational records.	Need-to-know, encryption, restricted export.
RESTRICTED	Facility layouts, configurations, identities, routes and security evidence.	Strong access approval, logging and controlled sharing.
HIGHLY RESTRICTED	Combined physical/digital paths, secrets, live control authority.	Segregated storage, minimal access and sanitised reporting.
LEGAL HOLD	Any evidence under preservation obligation.	No deletion or unapproved alteration.

42. Security and privacy requirements

EVM-SEC-001 Collect only evidence necessary for the authorised assurance purpose.

EVM-SEC-002 Secrets and reusable credentials shall not be stored as ordinary evidence values.

EVM-SEC-003 Evidence access, export, redaction and deletion shall be auditable for restricted classes.

EVM-SEC-004 Personal data shall be minimised, lawfully handled and retained only as justified.

EVM-SEC-005 Cross-tenant and cross-client evidence shall remain isolated.

EVM-SEC-006 Reports may abstract sensitive details while retaining stable traceability to protected evidence.

EVM-SEC-007 Evidence used across jurisdictions shall identify location, transfer and access restrictions.

Part XVIII - Graph-native and machine-readable model

43. Core graph nodes

Node	Purpose
EvidenceObject	Addressable source item.
EvidenceSet	Assertion-specific collection.
EvidencePackage	Control/system/gate package.
EvidencePortfolio	Capability-level evidence collection.
EvidenceRequirement	Required class and threshold.
EvidenceAssessment	Quality and sufficiency judgment.
EvidenceSource	Originating system/person/provider.
CollectionActivity	Acquisition event.
TransformationActivity	Derived evidence process.
CustodyEvent	Access/transfer/change event.
Assertion	Proposition supported/challenged.
UnknownRecord	Missing or unresolved evidence condition with type and severity.
EvidenceConflict	Contradictory evidence relationship.
ReviewRecord	Human challenge/approval.
Decision	Bounded Assurance Decision.

The canonical object classes for exchange are those of AIDCF-014 §7; the nodes above are the evidence-model view and map onto those classes.

44. Core graph relationships

Predicate	Meaning	Canonical AIDCF-014 §9 relationship
SUPPORTS	Evidence supports a precise assertion.	SUPPORTS
CHALLENGES	Evidence weakens or contradicts assertion.	CHALLENGES
QUALIFIES	Evidence limits scope/conditions.	QUALIFIES

Predicate	Meaning	Canonical AIDCF-014 §9 relationship
SATISFIES_REQUIREMENT	Evidence package meets requirement.	Evidence-model extension; exchanged as EVIDENCED_BY on the requirement with an assessment result.
COLLECTED_BY / GENERATED_BY / DERIVED_FROM	Provenance of an object.	Evidence-model extension (PROV-DM aligned).
OBSERVES / COVERS	Evidence records an entity/state, or covers scope, state, period or population.	Evidence-model extension.
VALIDATES	Test evidence evaluates a safeguard or claim.	SUPPORTS or CHALLENGES with claim type EFFECTIVENESS/RESILIENCE.
CONFLICTS_WITH	Objects/assertions materially disagree.	CHALLENGES plus EvidenceConflict.
SUPERSEDES	New object replaces current use.	SUPERSEDES
MEMBER_OF / CUSTODIAN_OF	Package membership; custody.	Evidence-model extension.
LIMITS	UNKNOWN/limitation constrains conclusion.	CONSTRAINS
EVIDENCES_GATE	Package contributes to Gate Result.	EVIDENCED_BY (Gate Result EVIDENCED_BY package)

45. Publication package

File	Purpose
aidcf-evm-v1.1.owl / .ttl	Formal evidence ontology and RDF/Turtle serialisation (namespace https://w3id.org/aidcf/ontology#; w3id registration pending).
aidcf-evm-context-v1.1.jsonld	JSON-LD context.
evidence-object-schema-v1.1.json	EvidenceObject exchange schema.
evidence-package-schema-v1.1.json	Package/portfolio schema.
aidcf-evm-shapes-v1.1.ttl	SHACL validation shapes.
evidence-types-v1.1.csv	Type registry (Appendix A).
quality-ratings-v1.1.csv	Quality dimension anchors (Appendix B).
AIDCF-007A registry evidenceProfile	Control Evidence Requirement Profiles for all 360 controls (Appendix C); no separate profile matrix is maintained in this artifact.
registries/gates/critical-gates.csv and critical-gate-criteria.csv	Gate evidence profile and criteria register (Appendix D).
retention-classes-v1.1.csv	Retention definitions.
conformance-tests/	Positive, negative, migration and security fixtures.
CHANGELOG.md / MIGRATION.md	Version history and transition guidance, including the Annex G requirement mapping.

Part XIX - Conformance, validation and nonconformance

46. Validation rules

Rule	Validation statement
EVM-VAL-001	Every EvidenceObject has stable ID, class, source, scope, observation time, classification and review state.
EVM-VAL-002	Every accepted material EvidenceObject links to at least one assertion or requirement, with subject and period.
EVM-VAL-003	Every derived object links to sources and transformation activity.
EVM-VAL-004	Every quality rating has rationale and reviewer.
EVM-VAL-005	Every evidence package evaluates timeliness, coverage, conflict and sufficiency.
EVM-VAL-006	Every Gate Result references its evidence requirement and package.
EVM-VAL-007	Every missing mandatory item creates an UNKNOWN Record or an explicit adverse result.
EVM-VAL-008	No UNKNOWN is encoded as favourable pass or numeric zero.
EVM-VAL-009	No evidence item is treated as universal proof across unreviewed assertions.
EVM-VAL-010	Every stale/superseded object remains historically traceable.
EVM-VAL-011	Every restricted item has handling and access metadata.
EVM-VAL-012	Every AI-generated assertion remains candidate until approved.
EVM-VAL-013	Every evidence deletion has authorised destruction record unless prohibited by hold.
EVM-VAL-014	Every test evidence object records initial state, method, conditions and anomalies.
EVM-VAL-015	Every Assurance Decision links to an immutable evidence manifest or snapshot.
EVM-VAL-016	Every UNKNOWN Record has a type U1–U10 and a severity US1–US5 with the determining test recorded.
EVM-VAL-017	Every control Evidence Package cites the AIDCF-007A registry version of the Evidence Requirement Profile applied.
EVM-VAL-018	Confidence records use CONF-0 to CONF-4 and do not exceed the ceilings in §27.

47. Evidence nonconformance classes

Class	Examples	Decision treatment
Critical	Fabricated/falsified evidence; concealed material conflict; unsafe unauthorised collection; custody breach affecting critical claim.	Stop use, preserve records, escalate, independently investigate and suspend affected decision (CG-11 FAIL in AIDCF-010).
Major	Missing provenance; unsupported effectiveness; stale critical evidence; incomplete gate package; hidden UNKNOWN.	Reject/limit evidence and reassess conclusion.
Minor	Non-material metadata defect or formatting inconsistency.	Correct and record.
Observation	Improvement opportunity without current decision impact.	Track through quality process.

Part XX - Worked evidence packages

48. Power continuity package

Component	Example
Assertions	Required power continuity and resilient capacity exist for declared load and states (claim types OPERATION and RESILIENCE).
E1	SLD, design basis, protection/coordination and capacity study, switching and fuel procedures.
E2	Protection settings, UPS/generator/EPMS configuration, asset inventory.
E3	Load, runtime, fuel, alarms, maintenance and incident evidence.
E4	Generator start/transfer, UPS runtime, load-step and restoration tests.
E5	Independent engineering/witness review where required.
Quality focus	Current load/configuration (T), representative states (C), shared dependencies, test safety.
Gate	CG-05; evidence floor S5.
UNKNOWN example	Unverified shared bus or fuel replenishment autonomy: U10/U1, graded US4 because fuel autonomy and replenishment is a CG-05 gate criterion (GC-05-5).
Decision limit	No resilience conclusion beyond tested load, duration and states.

49. Liquid cooling package

Component	Example
Assertions	Heat removal remains within envelope under NORMAL and defined DEGRADED states.
E1	P&ID, thermal model, design limits, water/coolant specification.
E2	CDU, controller, setpoint, sensor and loop configuration.
E3	Flow, temperature, pressure, leak, quality and alarm telemetry.
E4	Failover, isolation, leak response and thermal recovery test.
E5	Specialist engineering review for high-consequence or novel architecture.
Quality focus	Sensor trust, seasonal condition, transient response and shared loop dependencies.
Gate	CG-06; evidence floor S5.

Component	Example
UNKNOWN example	Secondary-loop autonomy not verified at peak load: U5 coverage gap, graded US4 because it affects gate criterion GC-06-3 (secondary-loop envelope at peak load).
Decision limit	No generalisation from steady low load to full transient AI demand.

50. Compute-storage-network package

Component	Example
Assertions	Required usable performance, integrity and recovery are sustained for defined Service Outcomes.
E1	Cluster/fabric/storage architecture and compatibility basis.
E2	Firmware, driver, topology, scheduler and storage configurations.
E3	Accelerator errors, fabric congestion, storage performance, capacity and incident records.
E4	Benchmark, node/fabric failure, checkpoint and restore tests.
E5	Independent validation where required.
Quality focus	Representative workload, topology version, error gaps and failure-domain coverage.
Gate	CG-08 and CG-10.
UNKNOWN example	Checkpoint recovery not tested after correlated metadata/fabric disruption: U5, graded US5 because it affects both CG-08 and CG-10.
Decision limit	Performance evidence does not prove recovery or integrity unless directly assessed.

51. Digital authority over infrastructure

Component	Example
Assertions	Only approved identities can cause consequential infrastructure actions within stated conditions.
E1	Access policy, responsibility and remote-support process.
E2	Identity, role, permission, gateway and controller configuration.
E3	Authentication, session, command, change and revocation logs.
E4	Authorised access-path, command restriction and revocation tests.
E7	Integrated authority-path analysis using AIDCF-005 semantics.
Quality focus	Effective identity, inherited privilege, dormant routes, provider access and rollback.
Gate	CG-15.
UNKNOWN example	Remote OEM command scope cannot be independently verified: U2/U6, graded US4.
Decision limit	Reachability alone does not prove authority or realised physical effect.

Appendix A - Evidence type registry

Each subtype is a controlled narrower term of its class. A capability profile may add narrower terms beneath a subtype without redefining the parent class.

Subtype ID	Class	Preferred label
E1-01	E1	Policy
E1-02	E1	Standard
E1-03	E1	Procedure
E1-04	E1	Runbook
E1-05	E1	Design Basis
E1-06	E1	Architecture Diagram
E1-07	E1	Single-Line Diagram
E1-08	E1	P&ID
E1-09	E1	As-Built Drawing
E1-10	E1	Contract
E1-11	E1	SLA
E1-12	E1	Attestation
E1-13	E1	Inventory
E1-14	E1	Risk Record
E2-01	E2	Device Configuration
E2-02	E2	Protection Setting
E2-03	E2	Firmware Baseline
E2-04	E2	Driver Matrix
E2-05	E2	Cluster Manifest
E2-06	E2	Scheduler Policy
E2-07	E2	Access Policy
E2-08	E2	Topology Export
E2-09	E2	Controller Logic
E2-10	E2	Configuration Snapshot
E2-11	E2	Golden Image
E3-01	E3	Telemetry Series
E3-02	E3	Event Log
E3-03	E3	Alarm Record
E3-04	E3	Capacity Metric
E3-05	E3	Performance Metric
E3-06	E3	Work Order
E3-07	E3	Maintenance Record
E3-08	E3	Incident Record
E3-09	E3	Change Record
E3-10	E3	Access Log
E3-11	E3	Video/Photo Observation
E3-12	E3	Inspection Record
E4-01	E4	Commissioning Test

Subtype ID	Class	Preferred label
E4-02	E4	Functional Test
E4-03	E4	Load Test
E4-04	E4	Failover Test
E4-05	E4	Recovery Test
E4-06	E4	Scenario Exercise
E4-07	E4	Simulation
E4-08	E4	Calibration Result
E4-09	E4	Witnessed Demonstration
E4-10	E4	Penetration/Control Test when authorised
E5-01	E5	Independent Engineering Review
E5-02	E5	Third-Party Assessment
E5-03	E5	Audit Report
E5-04	E5	Accredited Certificate
E5-05	E5	Laboratory Result
E5-06	E5	Provider Assurance Report
E5-07	E5	Regulatory Inspection
E5-08	E5	Independent Witness Record
E6-01	E6	Interview Record
E6-02	E6	Workshop Record
E6-03	E6	Walkthrough
E6-04	E6	Operator Demonstration
E6-05	E6	Expert Judgment
E6-06	E6	Shift Observation
E6-07	E6	Competency Record
E6-08	E6	Training Record
E7-01	E7	Trend Analysis
E7-02	E7	Correlation Analysis
E7-03	E7	Capacity Model
E7-04	E7	Failure Analysis
E7-05	E7	Dependency Graph
E7-06	E7	Reconciled Inventory
E7-07	E7	Evidence Quality Assessment
E7-08	E7	Score/Rating Output
E7-09	E7	AI-Assisted Extraction
E7-10	E7	Query Result

Appendix B - Quality rating matrix

Compact view of the anchors in §17 for tooling and rating sheets.

Rating	P Provenance	F Fitness	C Coverage	R Reliability	T Timeliness	I Integrity
0	Unknown origin	Not established / unlinked	Unknown coverage	Unassessed	Unknown age or state	Unknown integrity
1	Identified, uncontrolled source	Indirect subject	Anecdotal or isolated	Weak/uncontrolled method	Obsolete for represented state	Basic repository controls
2	Accountable internal source	Partial: other claim type, version or state	Limited sample/state	Plausible, lightly corroborated	Stale or change-sensitive	Verified source copy
3	Controlled authoritative source	Relevant and linked to assertion	Representative core scope	Controlled repeatable method	Acceptable for decision	Hash verified
4	Corroborated authoritative source	Specific to criterion, version and states	Broad scope and relevant states	Strong, independently corroborated	Current and aligned to baseline	Signature/timestamp verified
5	Independently verified origin and lineage	Discriminating for the criterion	Comprehensive population, period and states	Validated method, independent verification	Near-real-time, healthy collection	Tamper-evident end-to-end custody

Appendix C - Control Evidence Requirement Profile register

The 360 Evidence Requirement Profiles are carried in the AIDCF-007A Canonical Master Control Registry (evidenceProfile of each control record; CSV exports flatten array fields with ";" separators). The register uses exactly the thirteen fields in §31: claimType, profileTier, requiredClasses, optionalClasses, qualityFloor, sufficiencyFloor, stateCoverage, periodCoverage, independence, mandatoryItems, acceptableAlternatives, unknownEffect, reassessmentTriggers.

Register validation rules:

- claimType is consistent with requiredAssuranceLevel (DESIGN–AL2, IMPLEMENTATION–AL3, OPERATION–AL4, EFFECTIVENESS/RESILIENCE–AL5).
- requiredClasses satisfy the claim-type minimum pattern in §29 and the tier pattern in §32.
- qualityFloor names all six dimensions; sufficiencyFloor is S2–S5 and is at least S4 for CP-4 and S5 (or a documented strong floor) for CP-5.
- stateCoverage uses canonical operating states only and includes every state in which the control is claimed to operate.
- unknownEffect states a US severity; it is at least US4 where the control has a CRITICAL gate mapping (§31).

- mandatoryItems are specific to the control; no profile reuses another's mandatoryItems verbatim unless the evidence is genuinely identical.

Appendix D - Critical-gate evidence register

The normative gate evidence profiles are in Part XIII. The machine-readable register is registries/gates/critical-gates.csv (gateId, name, scope, evidenceFloor, requiredClasses, failCeiling, unknownCeiling, approvedNaConditions, ownerArtifact), and the gate criteria with their evidence and test methods are in registries/gates/critical-gate-criteria.csv. Gate criteria, roll-up and decision ceilings are owned by AIDCF-010.

Appendix E - Record schemas

E.1 EvidenceObject

Property	Cardinality	Type
evidenceId	1	Identifier (immutable)
title	1	String
description	0..1	String
evidenceClass	1	Controlled value E1–E7
subtype	0..1	Controlled value (Appendix A)
sourceId	1	Reference to EvidenceSource
owner	1	Reference to party/role
custodian	0..1	Reference to party/role
collector	0..1	Reference to agent
collectionMethod	0..1	Controlled value (assessment method code)
observationTime	1	Date-time or interval
collectionTime	0..1	Date-time
recordTime	0..1	Date-time
effectivePeriod	0..1	Interval
subjectRefs	0..*	References to subjects (assets, controls, gates, assertions)
scope	0..1	Structured scope (locations, population, operating states)
configurationVersion	0..1	Reference to baseline/version
integrityMethod	0..1	Controlled value
hash	0..*	Digest with algorithm
classification	1	Controlled value (§41)
retentionRule	0..1	Retention class (§21) with trigger
qualityAssessment	0..*	Quality vector per use (P/F/C/R/T/I with rationale)
reviewState	1	Controlled value (§9)
limitations	0..*	String

Property	Cardinality	Type
relatedEvidence	0..*	Typed reference (source, derivative, conflict, supersession, membership)

E.2 EvidenceAssessment

Property	Meaning
assessmentId	Stable ID.
assertionOrRequirement	Precise evaluation target.
evidenceRefs	Items assessed.
qualityVector	P/F/C/R/T/I ratings and rationale.
sufficiency	S0–S5 and rationale.
confidence	CONF-0 to CONF-4, flags and rationale.
conflictsAndUnknowns	Unresolved conditions with UNKNOWN type and severity.
reviewer/competence	Accountable reviewer.
date/snapshot	Temporal and version context.
decisionEffect	Result, Finding, gate or claim ceiling.

E.3 EvidenceManifest

Property	Meaning
manifestId/version	Immutable package identity.
package/portfolio	Parent scope.
evidenceItems	IDs, hashes, versions, locations and states.
missingItems	Expected but absent objects.
restrictedItems	Protected references and access rules.
supersession	Replaced objects and lineage.
signature/timestamp	Manifest integrity method.
review/approval	Release roles and date.

E.4 UnknownRecord

Property	Meaning
unknownId	Stable ID.
subject	Control assessment, gate criterion, dependency, boundary element or outcome affected.
type	U1–U10.
severity	US1–US5, the determining test, assessor and reviewer.
missingAttribute/reason	What is unknown and why.
potentialConsequence	Which result, gate, condition or decision could change.
evidenceNeeded	Evidence request reference.
owner/status/dates	Resolution owner, status and target.

Property	Meaning
disposition	Resolved, bounded by approved boundary change, or carried with decision ceiling (AIDCF-010 CG-12).
severityHistory	Prior severities, rationale and approver.

Appendix F - Review and release gates

Review role	Status	Focus
Methodology author	Draft completed	Evidence architecture coherence.
Assurance/evidence specialist	Pending	Quality, sufficiency, confidence and UNKNOWN severity model.
Electrical engineering	Pending	CG-05 evidence profile.
Mechanical/liquid cooling	Pending	CG-06 evidence profile.
Compute/storage/network	Pending	CG-08/CG-10 evidence profile.
Operations/reliability	Pending	Operational and recovery evidence.
OT/cybersecurity	Pending	Control-plane, identity and logging evidence.
Ontology/graph	Pending	Evidence classes/relationships and AIDCF-005/AIDCF-014 alignment.
Data/privacy/security	Pending	Classification, minimisation and handling.
Legal/records management	Pending	Retention, legal hold and disclosure.
Certification/accreditation	Pending	Evidence floors and scheme implications.
Independent practitioner	Pending	Usability and reproducibility.
Public release	Not approved	Mandatory review gates closed.

Acceptance criteria

- All AIDCF-007A controls carry a complete Evidence Requirement Profile in the registry that passes the Appendix C validation rules.
- Evidence quality dimensions remain visible and non-compensating where claim-critical.
- UNKNOWN, conflicts and stale evidence cannot be hidden by scoring; every UNKNOWN carries type and severity.
- Critical-gate evidence floors are enforceable.
- Human and machine-readable records preserve provenance and review.
- Security, privacy, retention and lawful handling are operationally usable.
- AIDCF-009 can directly consume the model for assessment procedures, and AIDCF-010 for results, gates and decisions.

Change record

Version	Status	Summary
0.1	Architecture draft	Evidence doctrine, hierarchy and taxonomy.

Version	Status	Summary
0.5	Integrated draft	Quality, sufficiency, gate, graph and lifecycle models.
1.0	Controlled-review draft	Full evidence standard and 360-control profile approach.
1.1	Draft for controlled review	Alignment to the canonical specification: AIDCF-007A, EC0–EC5, P/F/C/R/T/I meanings, confidence flags, UNKNOWN severity, registry-carried profiles, consecutive requirement IDs.

Annex G - Requirement identifier mapping (v1.0 to v1.1)

Requirement identifiers are numbered consecutively in v1.1. EVM-SEC and EVM-VAL identifiers are unchanged (EVM-VAL-016 to EVM-VAL-018 are new).

v1.0 ID	v1.1 ID
EVM-REQ-001	EVM-REQ-001
EVM-REQ-002	EVM-REQ-002
EVM-REQ-003	EVM-REQ-003
EVM-REQ-004	EVM-REQ-004
EVM-REQ-005	EVM-REQ-005
EVM-REQ-006	EVM-REQ-006
EVM-REQ-007	EVM-REQ-007
EVM-REQ-008	EVM-REQ-008
EVM-REQ-009	EVM-REQ-009
EVM-REQ-010	EVM-REQ-010
EVM-REQ-011	EVM-REQ-011
EVM-REQ-012	EVM-REQ-012
— (new in v1.1)	EVM-REQ-013
EVM-REQ-015	EVM-REQ-014
EVM-REQ-016	EVM-REQ-015
— (new in v1.1)	EVM-REQ-016
— (new in v1.1)	EVM-REQ-017
— (new in v1.1)	EVM-REQ-018
EVM-REQ-021	EVM-REQ-019
— (new in v1.1)	EVM-REQ-020
EVM-REQ-027	EVM-REQ-021
EVM-REQ-028	EVM-REQ-022
— (new in v1.1)	EVM-REQ-023
— (new in v1.1)	EVM-REQ-024
EVM-REQ-033	EVM-REQ-025
EVM-REQ-035	EVM-REQ-026
— (new in v1.1)	EVM-REQ-027

v1.0 ID	v1.1 ID
— (new in v1.1)	EVM-REQ-028
EVM-REQ-037	EVM-REQ-029

Final doctrine

Evidence is the currency of AIDCF assurance. An evidence item has no universal authority outside its provenance, fitness, scope, time, state, method and assertion. A strong-looking package cannot compensate for a missing mandatory class, critical coverage gap, unresolved conflict or failed gate. When evidence cannot support the requested conclusion, AIDCF preserves UNKNOWN, grades its severity, states the limitation and withholds stronger assurance rather than manufacturing certainty.